



COMPUTER Network

22CSE62 –M.5.1



Module 5 -Application layer and Security mechanisms

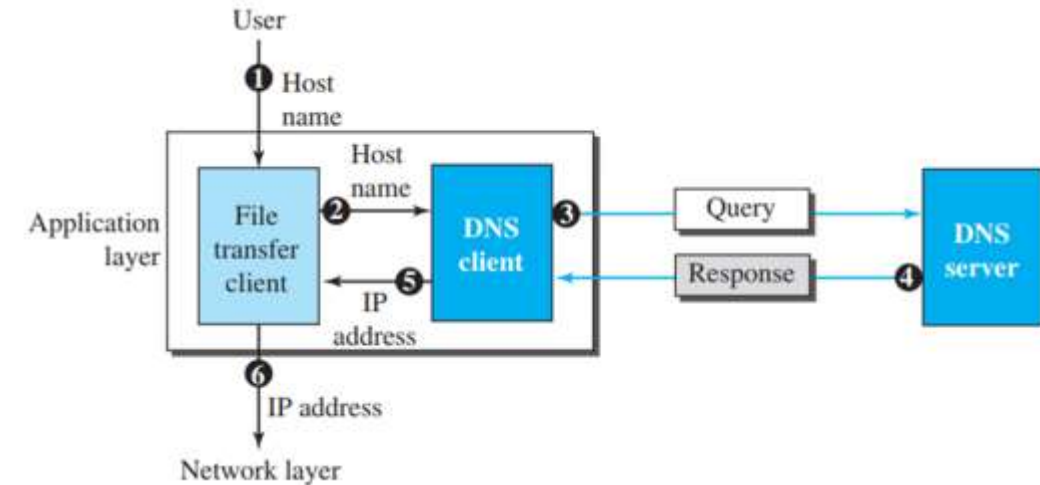


- **Overview of:**
 - Domain Name Space,
 - Remote Logging, Email, FTP, WWW, HTTP
 - Security – Network Layer Security, Application Layer Security



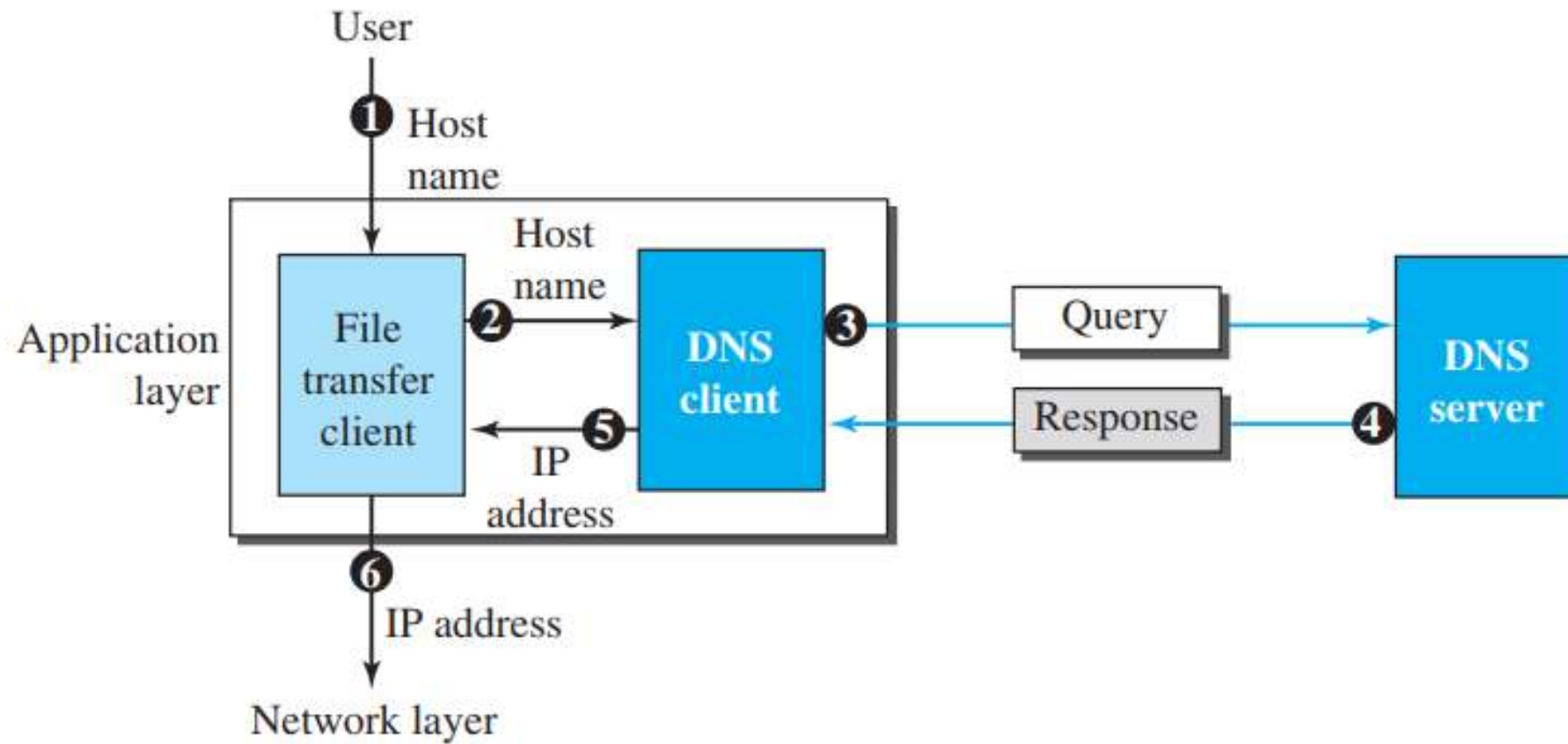
DOMAIN NAME SYSTEM (DNS)

- The internet model that follow the **client/server** paradigm.
- The **DNS is a supporting program** that is used by other programs such as E-mail.
- A user of a e-mail program may know the **e-mail address of the recipient**; however, the **IP protocol needs the IP address**.
- The DNS client program sends a request to a DNS server to map the e-mail address to the corresponding IP address.
- **To identify the remote system/user, TCP/IP protocols use the IP address, which uniquely identifies the connection of a host to the internet.**
- However, people prefer to use **names** instead of numeric values.
- The DNS system that **can map a name to an address** (or) address to a name.



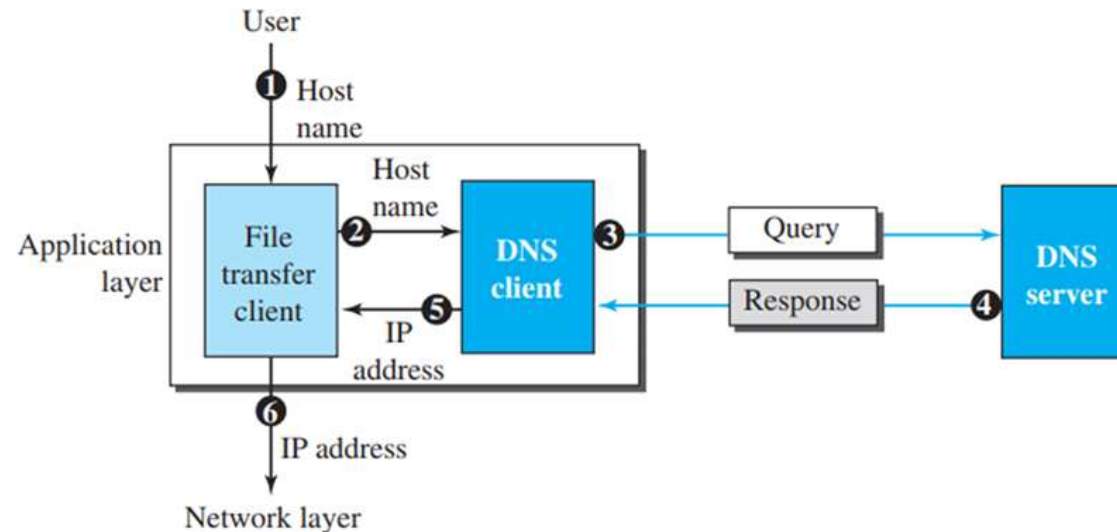


Purpose of DNS





1. The user passes the host name to the file transfer client.
2. The file transfer client passes the host name to the DNS client.
3. Each computer, after being booted, knows the address of one DNS server. The DNS client sends a message to a DNS server with a query that gives the file transfer server name using the known IP address of the DNS server.
4. The DNS server responds with the IP address of the desired file transfer server.
5. The DNS server passes the IP address to the file transfer client.
6. The file transfer client now uses the received IP address to access the file transfer server.





- When the internet was small, mapping was done by using a **host file**. [two columns-**names** and **address-host** store it-update periodic]
- Today it is impossible, bcoz the host file would be too large and updating problem.
- The solution is to maintain in **one computer** and allow **centralized access** [huge traffic]
- Huge information divided into small parts today and stored different computer. [host can contact the **closest computer** **holding the needed information**. [method used by DNS]



Name Space

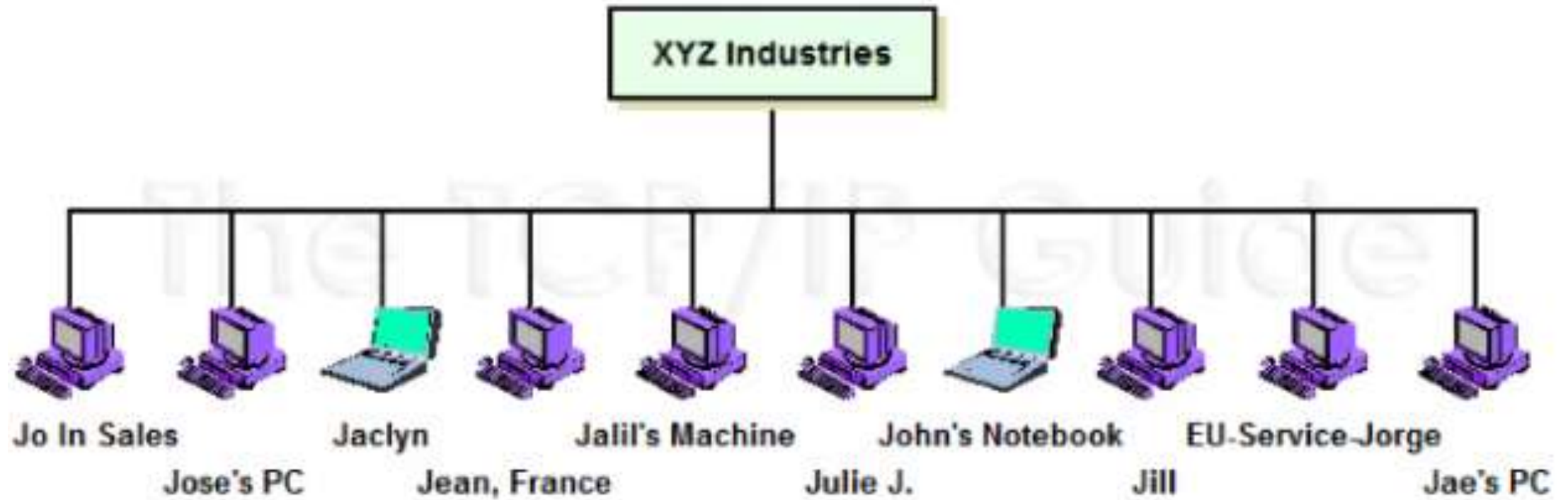
- It is unambiguous, the name assigned to machines must be unique.
- Name space map each address to a unique name in two ways.
 - Flat Name space
 - Hierarchical Name Space.

Flat Name Space:

- ✓ A name in this space is a sequence of characters without structure.
- ✓ A name may (or) may not have a common section.[it has no meaning].
- ✓ It cannot be used in internet.[duplication].



flat name space





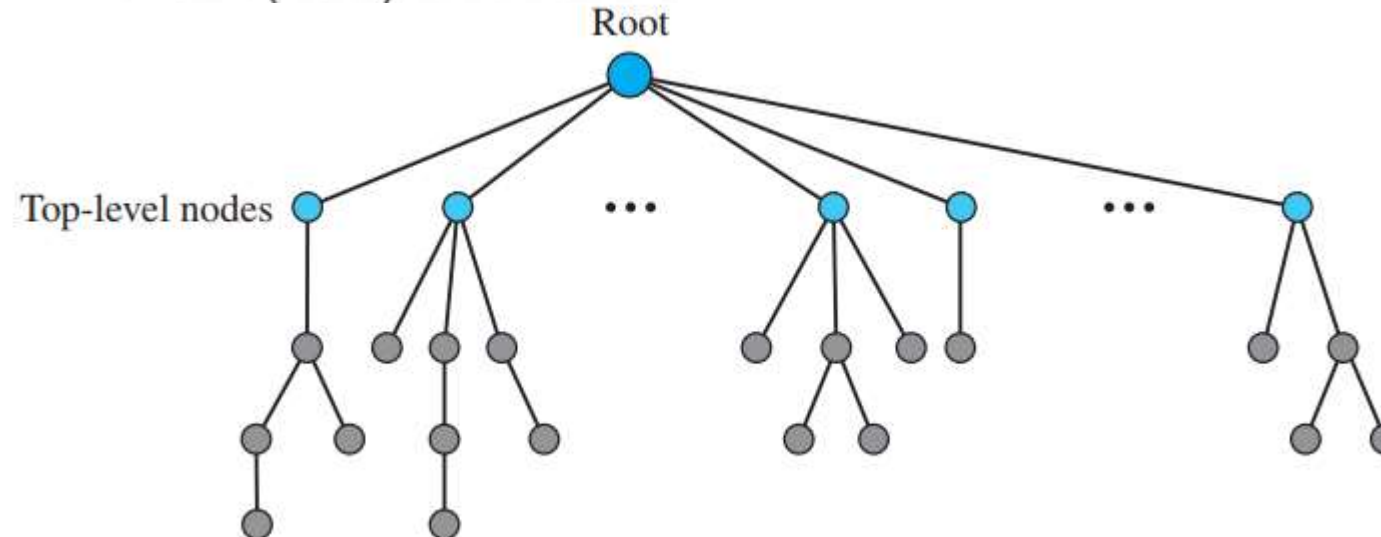
hierarchical name space

- Each name has several parts.
- The first part define the nature of the organization.
- The second part can define name of an organization.
- The third part can define departments in the organization, and so on.
- The central authority assigned only the first two part the name space the rest of parts are assigned organization itself.
- The organization can add prefix(or) suffix to the name to define its host or resource.
- The organization **need not worry about the same name** chosen by the other management for their resource.



Domain Name Space

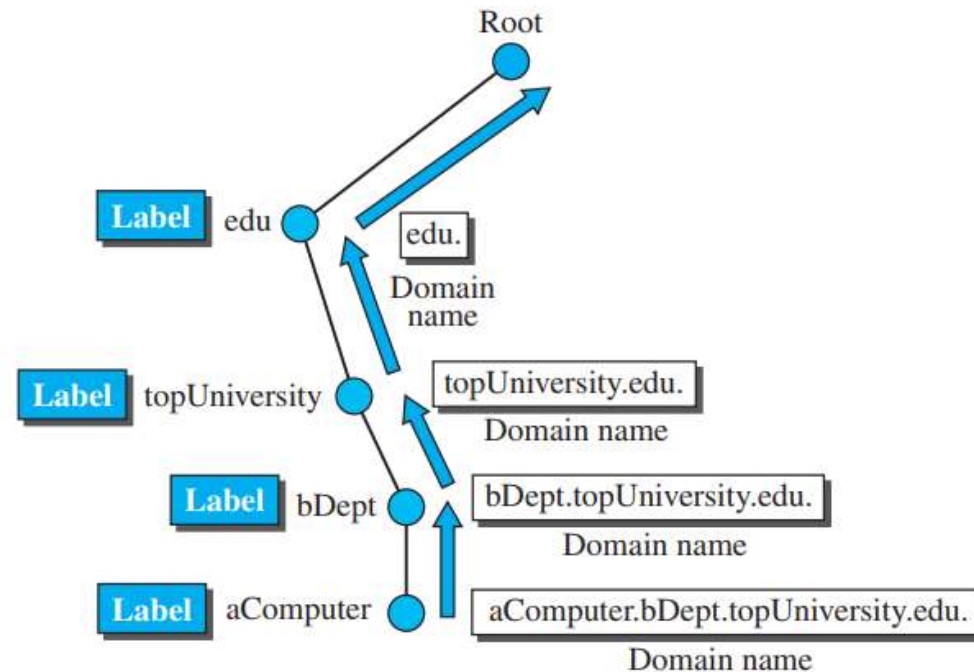
- When we have hierarchical name space, a domain name space to be designed.
- In that tree names are defined in an inverted-tree with one root at the top.
- The tree can have only 128 levels.
- Level 0(root) to level 127





Label

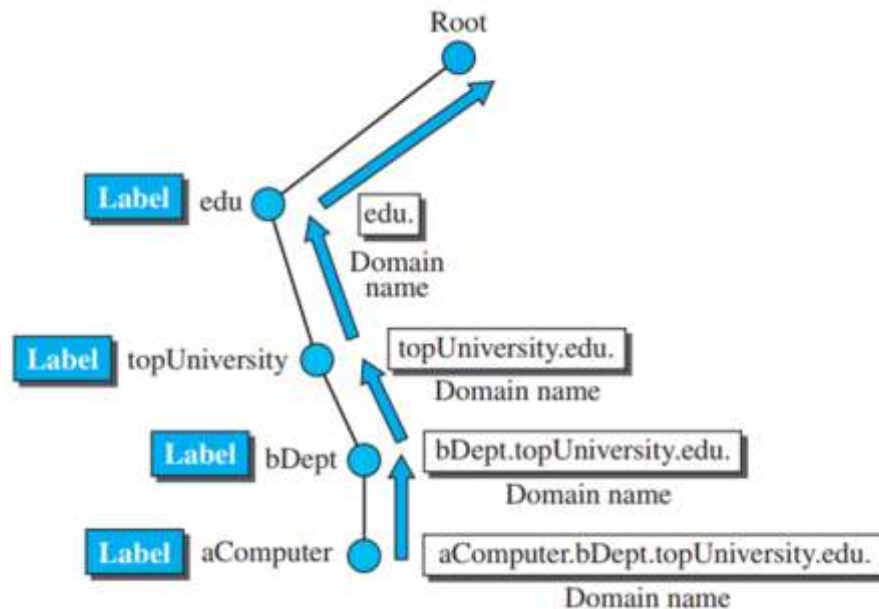
- Each node in the tree has a **label**, which is a string with a maximum of **63** characters.
- The root label is a **null string(empty)**.





Domain Name

- A full domain name is a **sequence of labels separated by dots.**
- The domain names are always **read** from the **node up to the root.**
- Finally, it end with null(root node)





fully qualified domain name (FQDN)

- A fully qualified domain name (FQDN) is the **complete domain name** for a specific computer, or host, on the Internet.
- The FQDN consists of two parts: the **hostname** and the **domain name**.
- If the label is terminated by a **null string(.)**, it is called a FQDN
- For example, an FQDN for a hypothetical mail server might be mymail.somecollege.edu.
- The hostname is mymail, and the host is located within the domain somecollege.edu.



partially qualified domain name (PQDN)

- If a label is not terminated by a **NULL string**, it is called a PQDN.
- It starts from a node, but it does not reach the root.
- Here the **resolver** can supply the missing part, called the **suffix**, to create an FQDN.
- Example:

FQDN and PQDN

- Google
- Yahoo
- Annauniv
- Kct

FQDN

challenger.atc.fhda.edu.
cs.hmme.com.
www.funny.int.

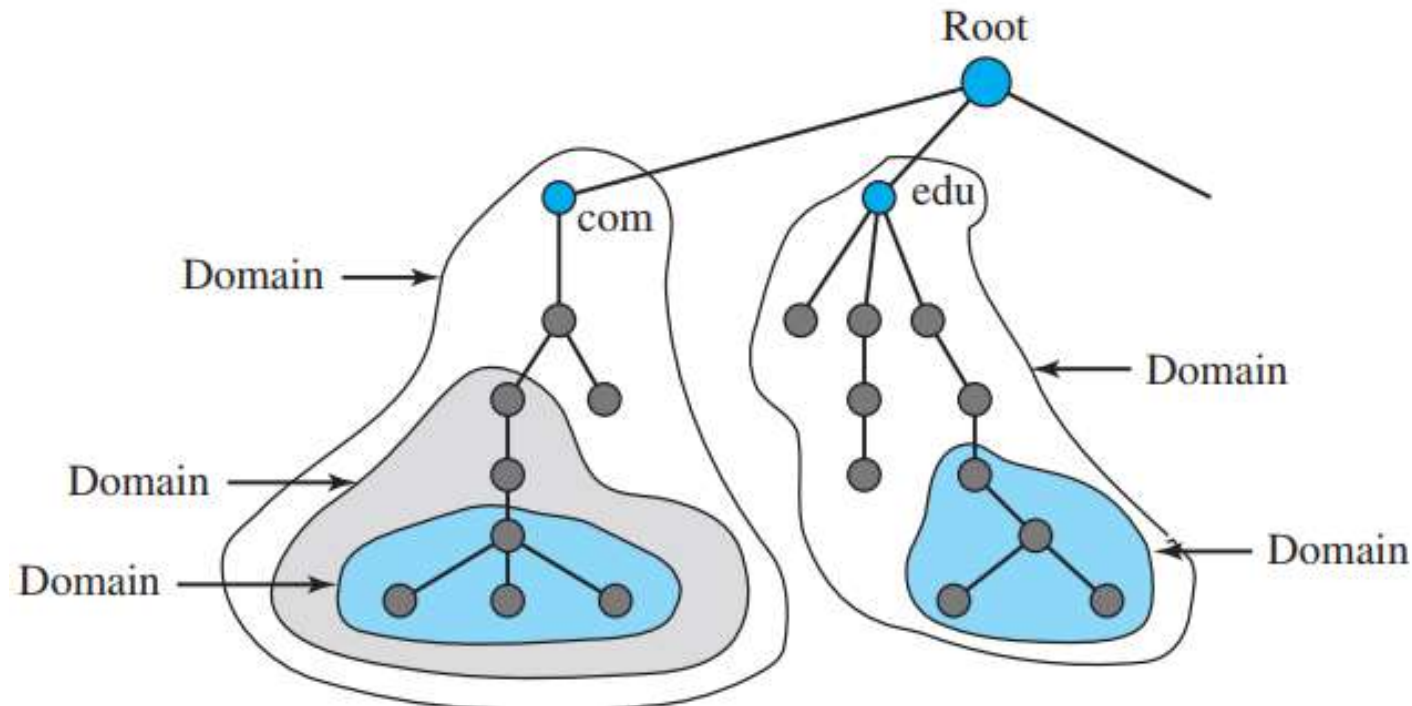
PQDN

challenger.atc.fhda.edu
cs.hmme
www



Domains

- A domain is a **subtree** of the domain name space.
- The name of the domain is the domain name of the node at the top of the subtree.





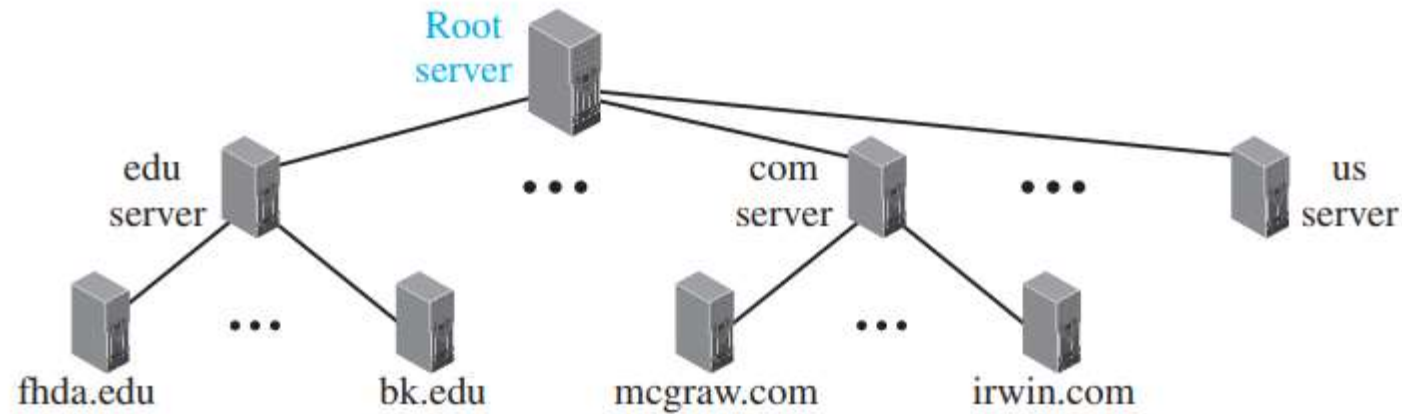
Distribution of Name Space

- The information contained in the domain name space must be stored.
- It is inefficient also unreliable[one computer store huge information.]



Hierarchy of Name Servers

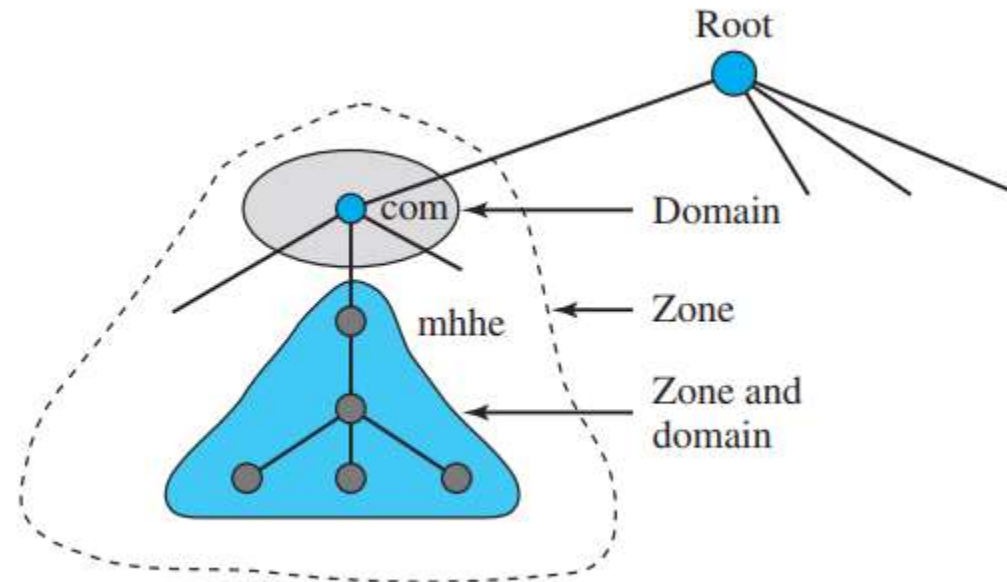
- The solution to these problems is to distribute the information among many computers called **DNS servers**.
- We create many sub DNS server based on the requirement[each divided into sub domain]





Zone

- When a server dedicated for (responsible) over is called a **zone**.





Root Server

- A root server is a server, whose **zone** consists of the whole tree.
- A root server usually **does not store any information** but **authority to other servers**.



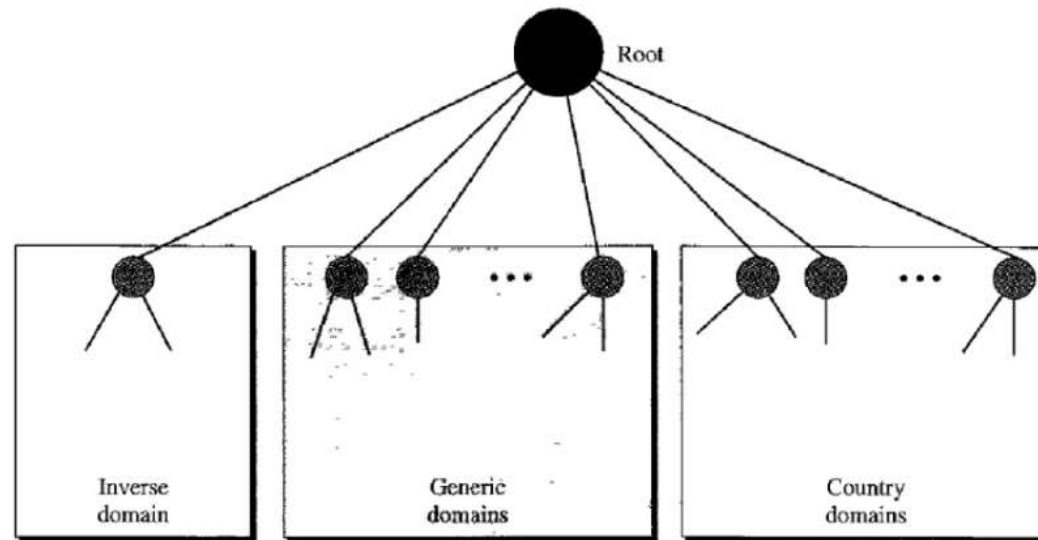
Primary and Secondary Servers

- DNS defines two types of servers:
- A **primary server** -stores a **file about the zone**, responsible for **creating , maintaining, and updating the zone file**.
- A **secondary server** – that transfers the complete information about a zone from another server and store the file on its local disk.



DNS in the Internet

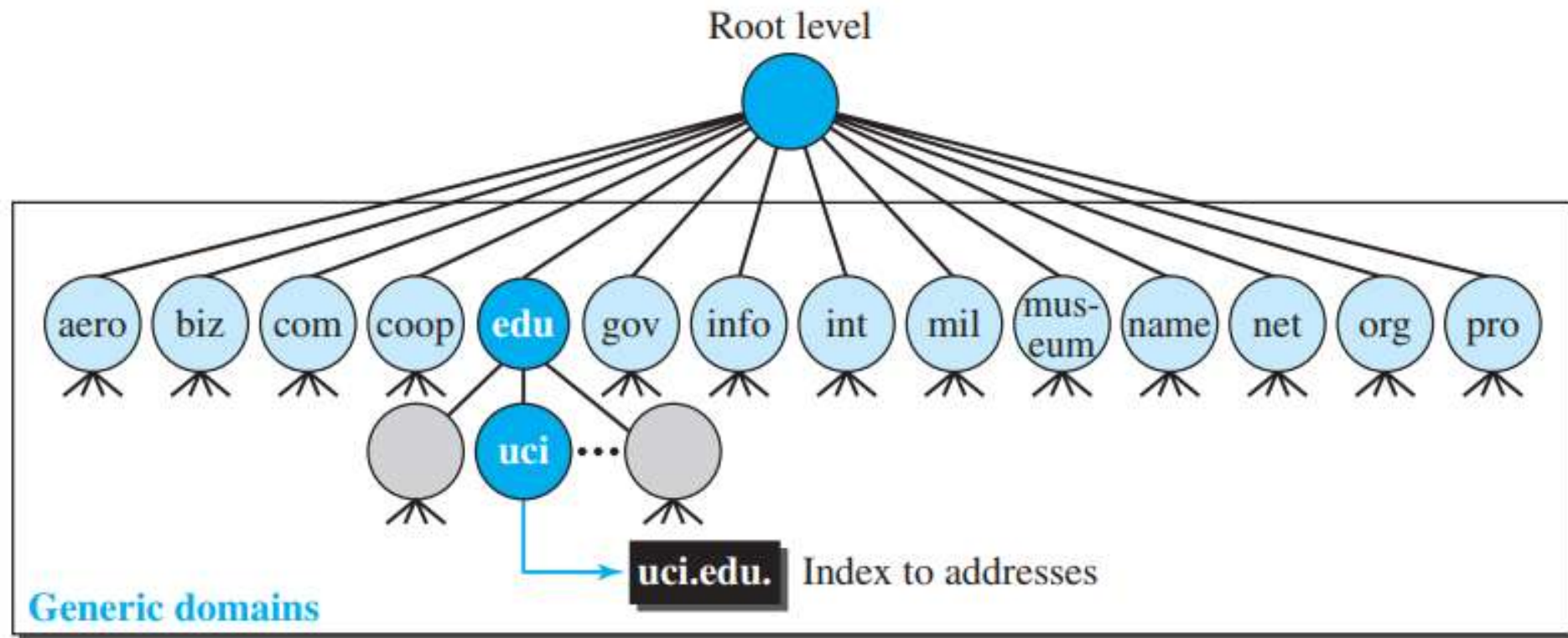
- In the internet, the domain space(tree) is divided into three different section:
 - » Generic domains
 - » Country domains
 - » Inverse domains





Generic Domains

- It define registered hosts according to their generic behaviour.



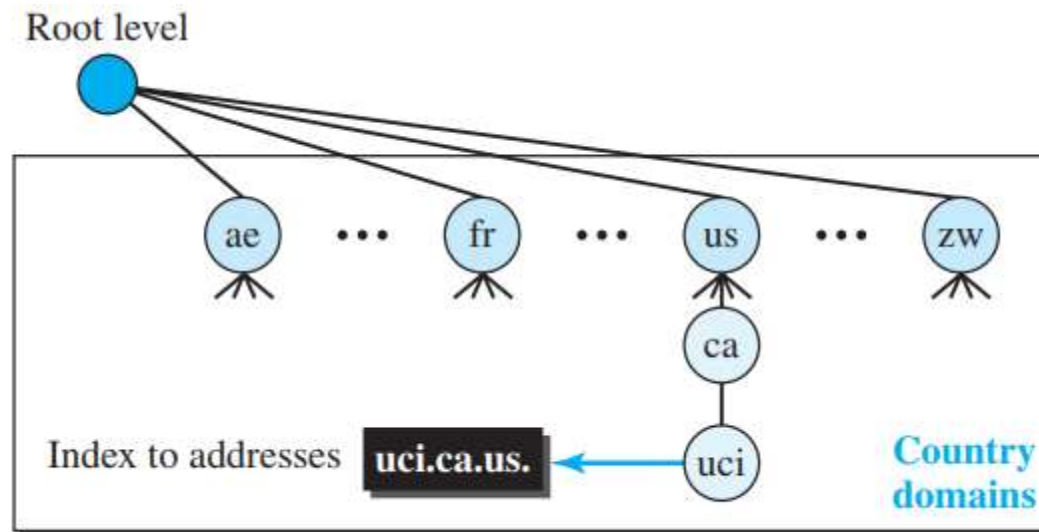


Generic domain labels

<i>Label</i>	<i>Description</i>	<i>Label</i>	<i>Description</i>
aero	Airlines and aerospace	int	International organizations
biz	Businesses or firms	mil	Military groups
com	Commercial organizations	museum	Museums
coop	Cooperative organizations	name	Personal names (individuals)
edu	Educational institutions	net	Network support centers
gov	Government institutions	org	Nonprofit organizations
info	Information service providers	pro	Professional organizations



Country Domains





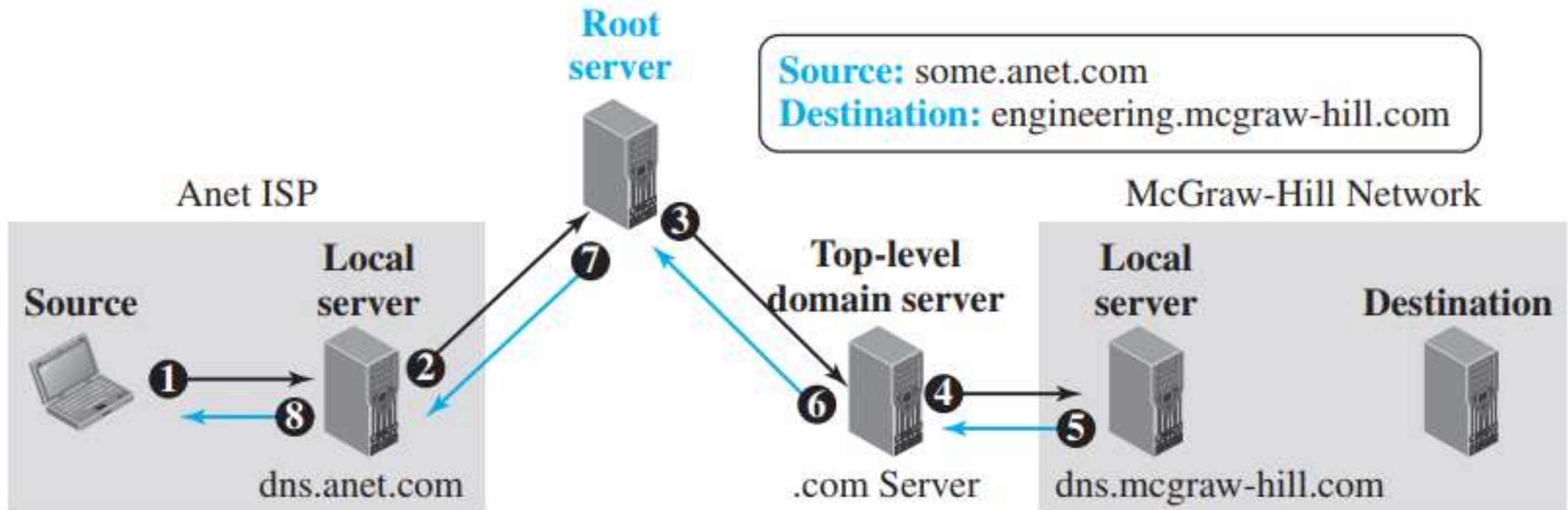
RESOLUTION

- Mapping a **name** to an address or **an address** to a name is called *name-address resolution*.
- DNS is designed as a **client/server** application.
- A host that **needs to map** an address to a name or a name to an address calls a DNS client called a **resolver**.
- The resolver accesses the **closest DNS server** with a mapping request.
- If the server has the information, it satisfies the resolver; otherwise, it either refers the resolver to other servers or asks other servers to provide the information.
- here are three methods of resolution:
 - Recursive
 - Iterative
 - Caching



Types of Resolution: Recursive Resolution

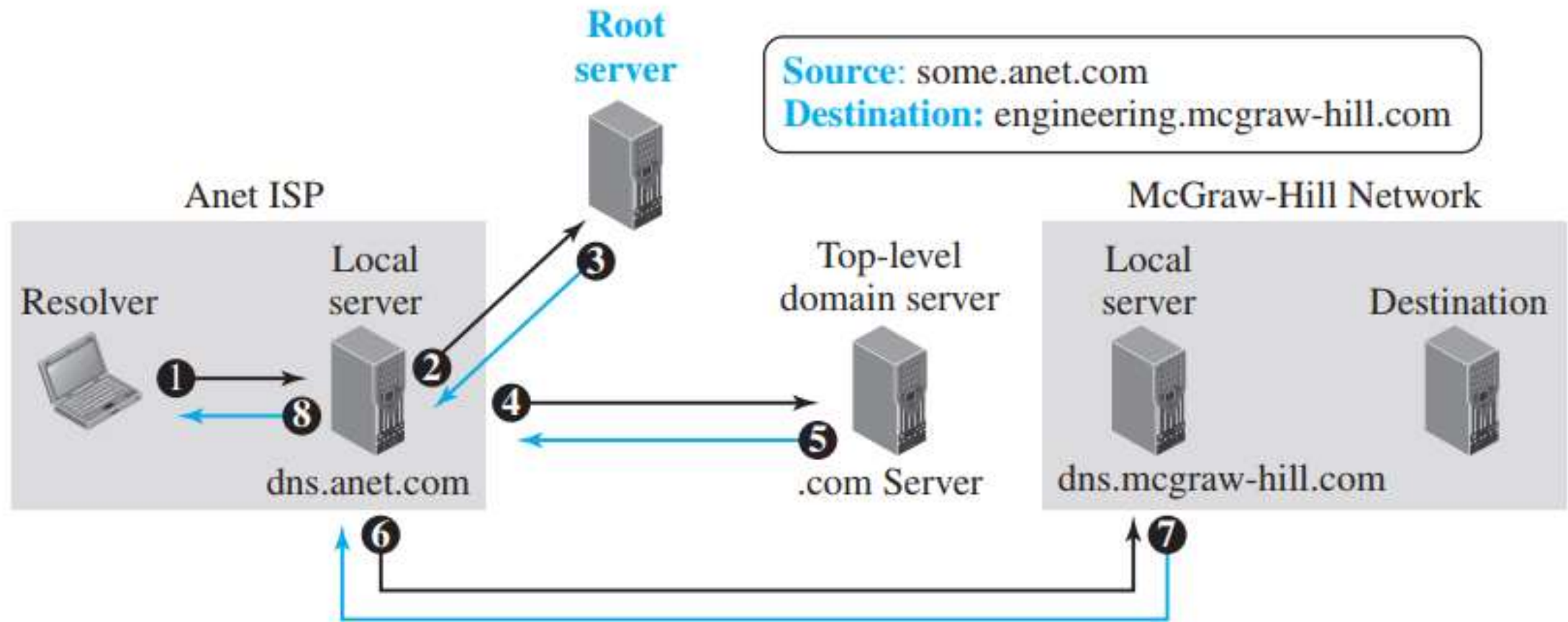
- The client (resolver) can ask for a recursive answer from a name server. This means that the resolver expects the server to supply the final answer.
 - If the server is the authority for the domain name, it checks its database and responds.
 - If the server is not the authority, it sends the request to another server (the parent usually) and waits for the response.
- If the parent is the authority, it responds; otherwise, it sends the query to yet another server.
- When the query is finally resolved, the response travels back until it finally reaches the requesting client. This is called recursive resolution





Types of Resolution: Iterative Resolution

- If the client does not ask for a recursive answer, the mapping can be done iteratively.
- If the server is an authority for the name, it sends the answer. If it is not, it returns (to the client) the IP address of the server that it thinks can resolve the query.
- The client is responsible for repeating the query to this second server. If the newly addressed server can resolve the problem, it answers the query with the IP address; otherwise, it returns the IP address of a new server to the client.
- Now the client must repeat the query to the third server.
- This process is called iterative resolution because the **client repeats the same query to multiple servers.**





Types of Resolution: CACHING

- Each time a server receives a query for a name that is not in its domain, it needs to search its database for a server IP address.
- Reduction of this search time would increase efficiency.
- DNS handles this with a mechanism called caching.
- When a server asks for a mapping from another server and receives the response, it stores this information in its cache memory before sending it to the client.
- If the same or another client asks for the same mapping, it can check its cache memory and solve the problem.

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TELENET

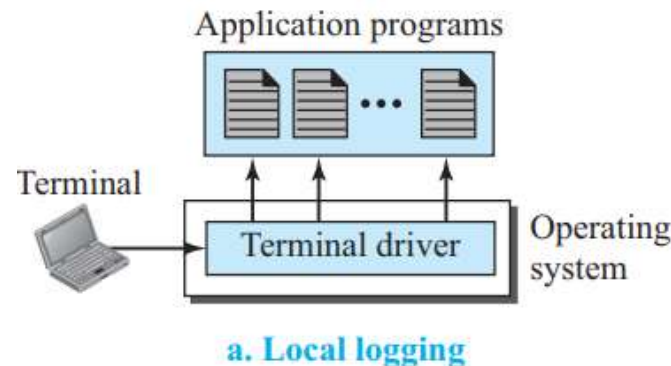
- Telnet (Terminal Networks) is a network protocol and software application used to access and manage remote computers over the Internet or within a TCP/IP network.
 - Telnet was developed in 1969 and later standardized as one of the earliest Internet protocols by the Internet Engineering Task Force (IETF).
- It allows users to run programs and **access data on remote systems**, especially within the same Local Area Network (LAN).
- The Telnet service typically operates **on port number 23** of the host machine.
- A key limitation of Telnet is that it **transmits data in plain text**, without encryption or strong security mechanisms.
- Due to these security vulnerabilities, Telnet has largely been replaced by **Secure Shell (SSH)**, which provides encrypted and secure communication.

• .



Local Login

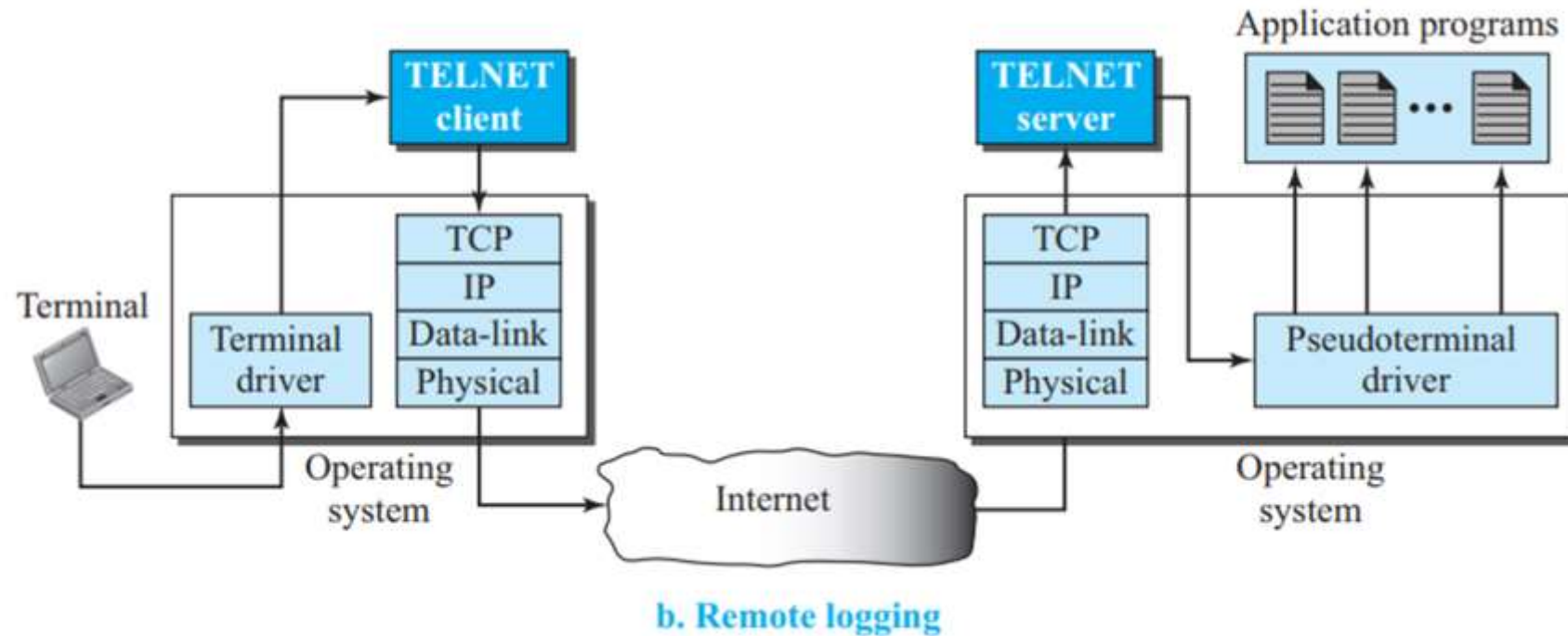
- When a user logs into their **own system** (local time-sharing system), it is called **local login**.
- The process works as follows:
 - The user types commands using a keyboard (terminal or terminal emulator).
 - These keystrokes are received by the **terminal driver**.
 - The terminal driver forwards the input to the **operating system (OS)**.
 - The OS interprets the commands and executes the required **application program or utility**.





Remote Logging

- When a user wants to access an application or utility on a remote machine, it is called remote login.
- In this process, both Telnet client and Telnet server are used.
-





1. User Input

- The user types commands on the local system (keyboard/terminal)

2. Terminal Driver

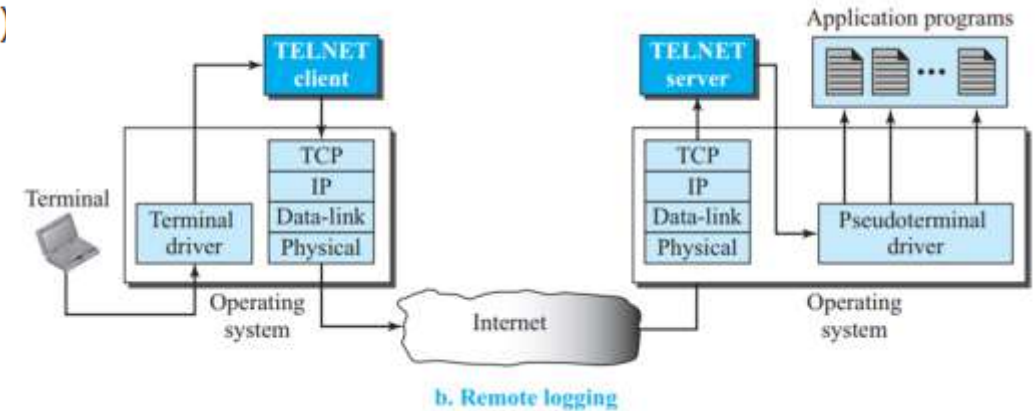
- The terminal driver accepts the keystrokes.
- The **local operating system does not interpret** these commands.

3. Telnet Client Processing

- The input is passed to the **Telnet client**.
- The client converts the characters into a standard format called:
👉 **Network Virtual Terminal (NVT)** format (universal character representation).

4. Transmission via TCP/IP

- The NVT data is sent through the **TCP/IP network** to the remote machine.





5. Telnet Server Processing

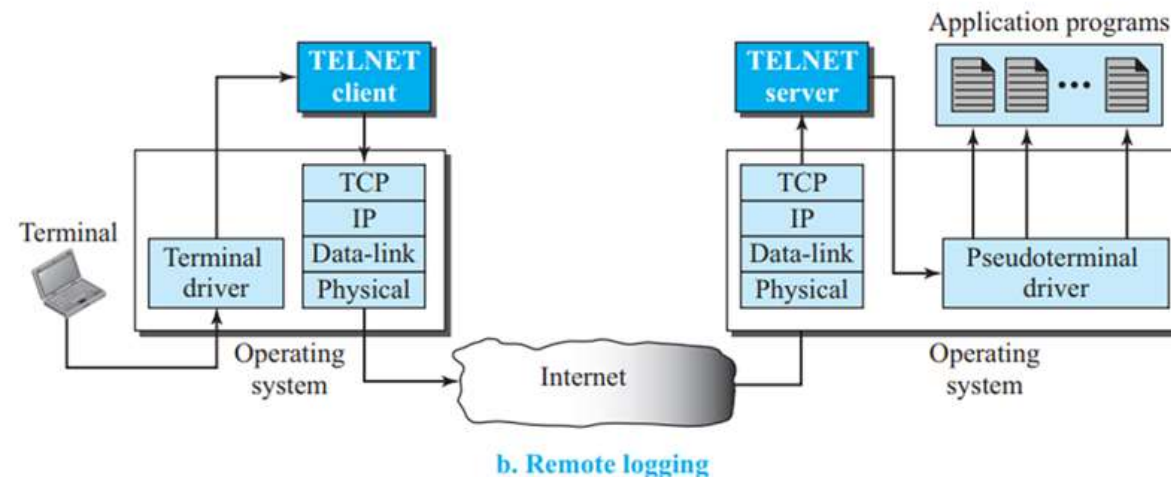
- The **Telnet server** on the remote system receives the data.
- It converts NVT format into the remote machine's native format.

6. Execution on Remote Machine

- The remote operating system interprets the command.
- The requested application/program is executed.

7. Response Back

- Output is sent back through the same process (server → client → user screen).





What is NVT (Network Virtual Terminal)?

- Different computers and operating systems use **different character representations and control commands**.
 - Example:
 - DOS uses **Ctrl + Z** for end-of-file
 - UNIX uses **Ctrl + D**

👉 This creates compatibility issues during remote communication.

Solution: NVT

- Telnet solves this problem using a **standard interface called NVT (Network Virtual Terminal)**.
 - It acts as a **universal language** between different systems.
1. The **Telnet client** converts user input (data/commands) into **NVT format**.
 2. The data is transmitted over the network.
 3. The **Telnet server** converts NVT data into the **remote system's native format**.
 4. The remote machine processes the command and sends back the response (again via NVT).



How NVT Works

1. The **Telnet client** converts user input (data/commands) into **NVT format**.
2. The data is transmitted over the network.
3. The **Telnet server** converts NVT data into the **remote system's native format**.
4. The remote machine processes the command and sends back the response (again via NVT).



How NVT Works

NVT Character Set

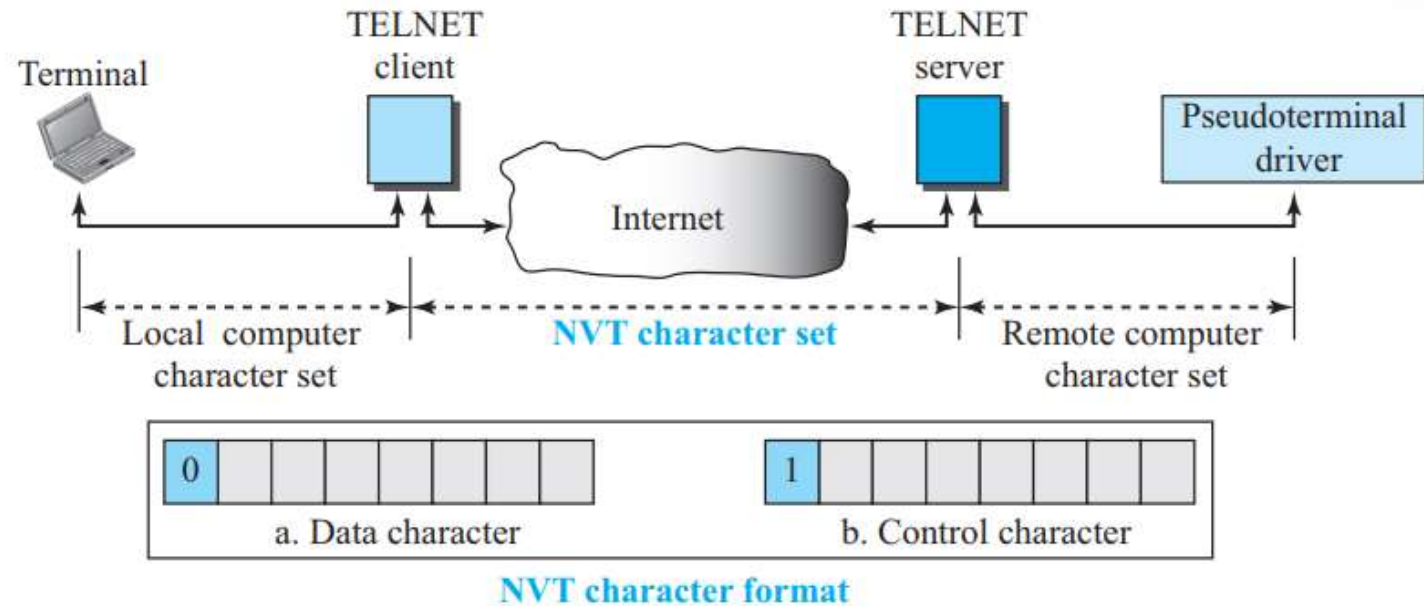
NVT uses two types of characters, both represented as 8-bit bytes:

1. Data Characters

- Uses NVT ASCII (American Standard Code for Information Interchange)
- Structure:
 - Lower 7 bits → Same as standard ASCII
 - Highest bit (8th bit) → Always 0

2. Control Characters

- Used for **control functions** (like commands, signals, etc.)
- Helps manage communication between client and server





Control Characters

To send control characters between computers (from client to server or vice versa), NVT uses an 8-bit character set in which the highest order bit is set to 1.

Lists some of the control characters and their meanings.

TELNET Options

- TELNET lets the client and server negotiate options before or during the use of the service.
- Options are extra features available to a user with a more sophisticated terminal.

Character	Decimal	Binary	Meaning
EOF	236	11101100	End of file
EOR	239	11101111	End of record
SE	240	11110000	Suboption end
NOP	241	11110001	No operation
DM	242	11110010	Data mark
BRK	243	11110011	Break
IP	244	11110100	Interrupt process
AO	245	11110101	Abort output
AYT	246	11110110	Are you there?
EC	247	11110111	Erase character
EL	248	11111000	Erase line
GA	249	11111001	Go ahead
SB	250	11111010	Suboption begin
WILL	251	11111011	Agreement to enable option
WONT	252	11111100	Refusal to enable option
DO	253	11111101	Approval to option request
DONT	254	11111110	Denial of option request
IAC	255	11111111	Interpret (the next character) as control

Code	Option	Meaning
0	Binary	Interpret as 8-bit binary transmission
1	Echo	Echo the data received on one side to the other
3	Suppress go-ahead	Suppress go-ahead signals after data
5	Status	Request the status of TELNET
6	Timing mark	Define the timing marks
24	Terminal type	Set the terminal type
32	Terminal speed	Set the terminal speed
34	Line mode	Change to line mode

ELECTRONIC MAIL





ELECTRONIC MAIL

- One of the most popular Internet services is electronic mail (e-mail).
- At the beginning of the Internet era, the messages sent by electronic mail were short and consisted of text only.
- Today, electronic mail is much more complex. It allows a message to include text, audio, and video.
- It also allows one message to be sent to one or more recipients



Format of an e-mail

Behrouz Forouzan
20122 Olive Street
Bellbury, CA 91000

William Shane
1400 Los Gatos Street
San Louis, CA 91005



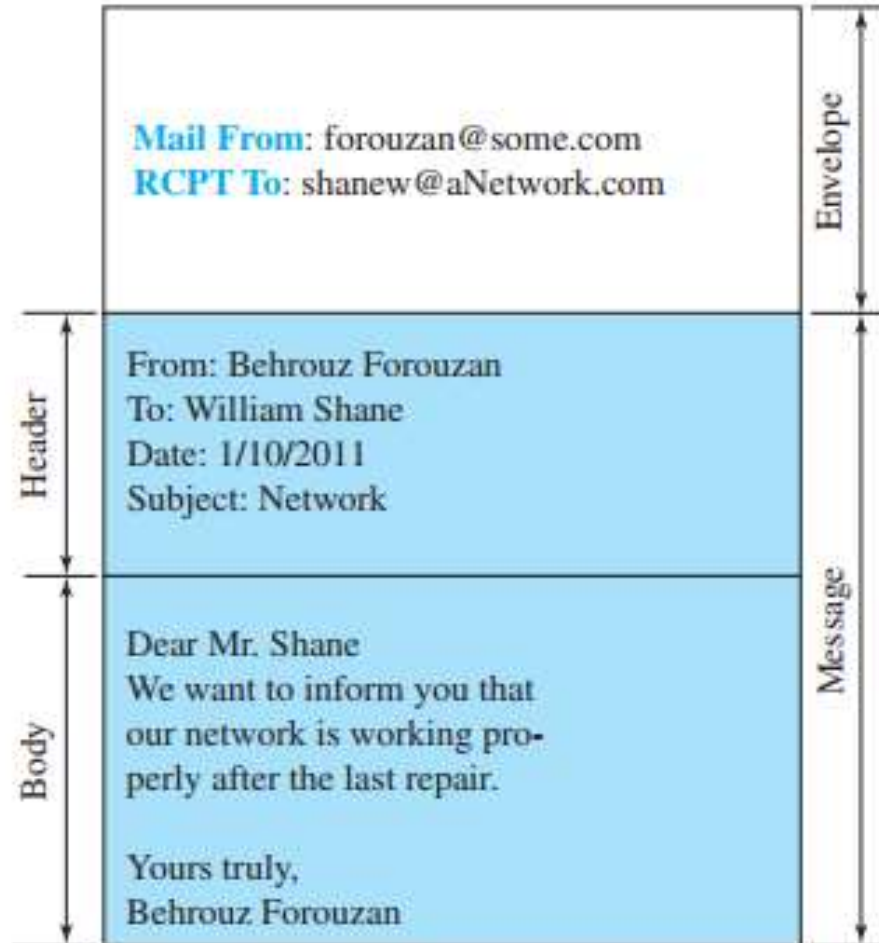
Behrouz Forouzan
20122 Olive Street
Bellbury, CA 91000
Jan. 10, 2011

Subject: Network

Dear Mr. Shane
We want to inform you that
our network is working properly
after the last repair.

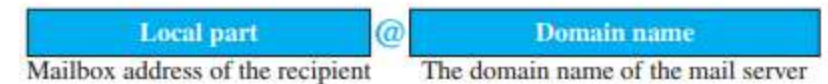
Yours truly,
Behrouz Forouzan

Postal mail



Electronic mail

E-mail address





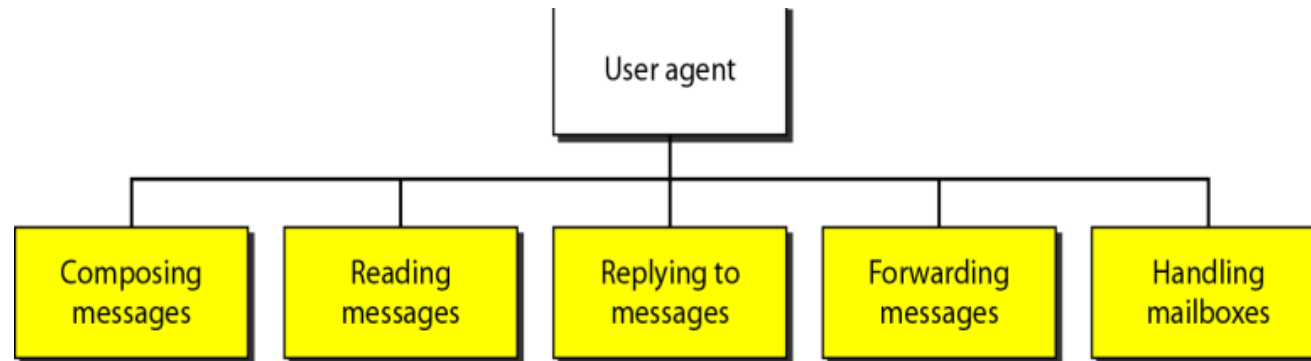
Components of e-mail

- An e-mail system is based on the three main components:
 - User Agent (UA)
 - Message Transfer Agent (MTA)
 - Message Access Agent (MAA).



User Agents(UA)

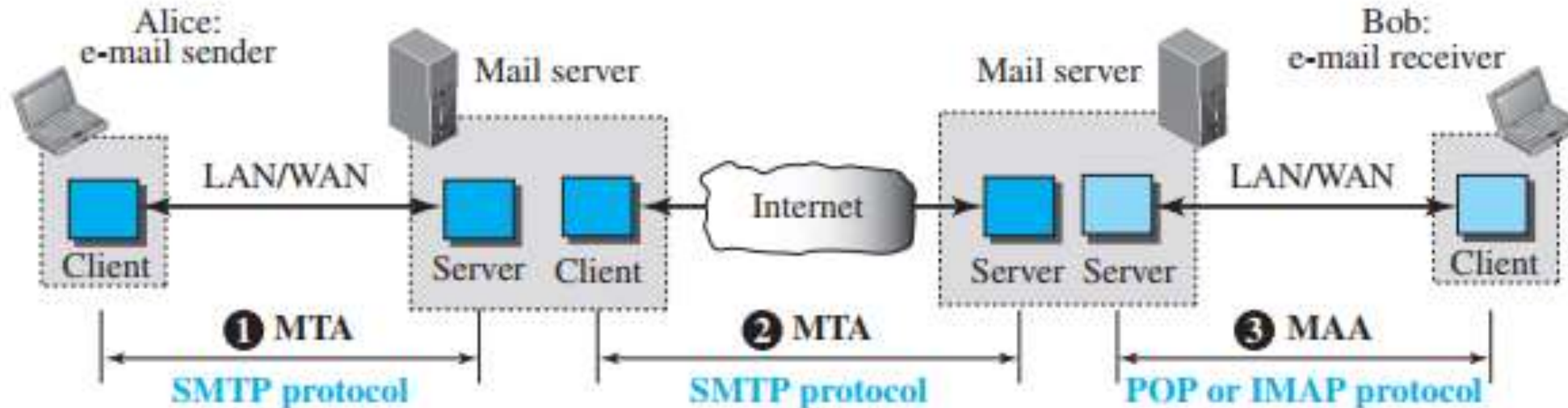
- The user agent (UA) provides services to the user to make the process of sending and receiving a message easier.
- A user agent is software (a software agent) that is acting on behalf of a user.
- There are two types of user agents: command-driven and GUI-based.





Message Transfer Agent (MTA): SMTP

- The actual mail transfer is done through message transfer agents.
- To send mail, a system must have the client MTA, and to receive mail, a system must have a server MTA.
- The formal protocol that defines the MTA client and server in the Internet is called the Simple Mail Transfer Protocol (SMTP).





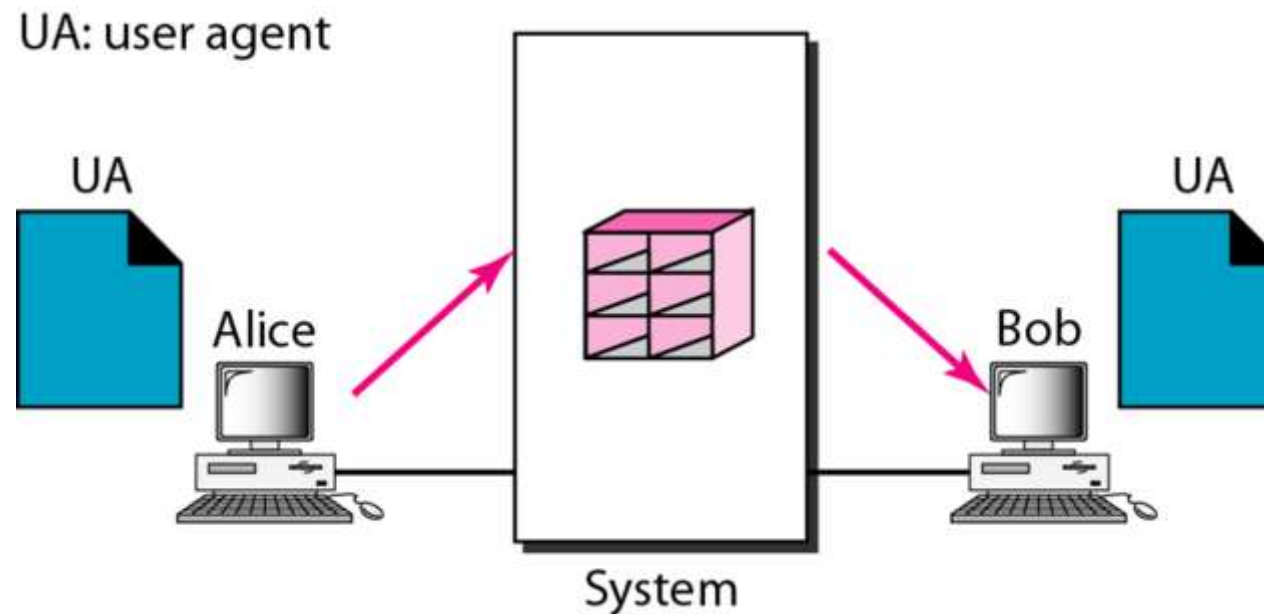
Architecture of e-mail

- Architecture of e-mail can be described using four scenarios
- We begin with the simplest situation and add complexity as we proceed.
- The fourth scenario is the most common in the exchange of email.



Architecture of e-mail - First Scenario:

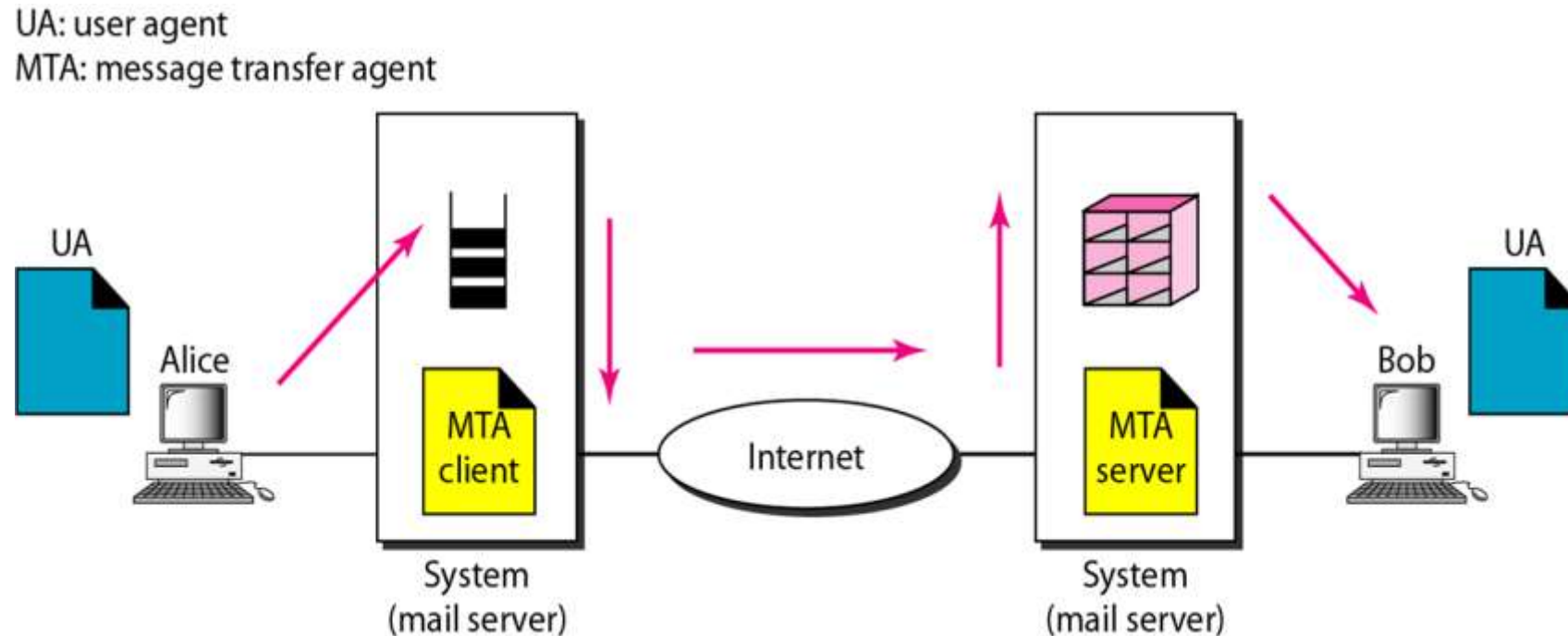
- When the sender and the receiver of an e-mail are on **the same system**, we need only two user agents. A mailbox is part of a local hard drive, a special file with permission restrictions. Only the owner of the mailbox has access to it.





Architecture of e-mail - Second Scenario:

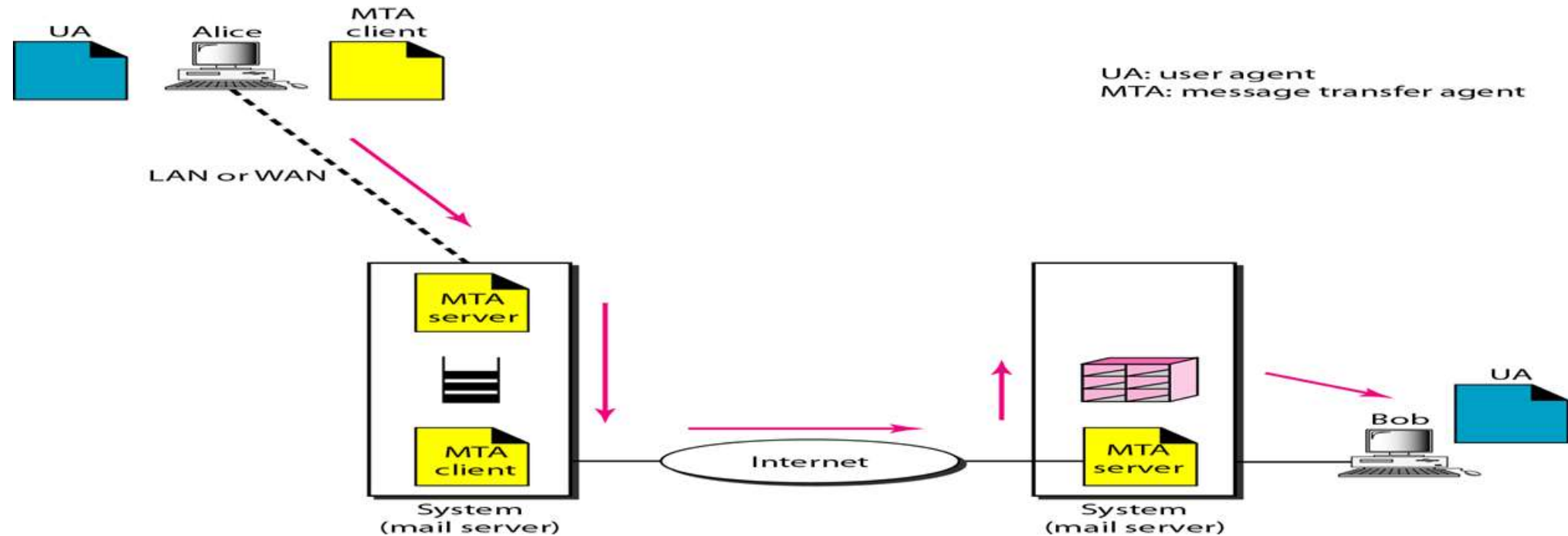
- When the sender and the receiver of an e-mail are on different systems, we need two UAs and a pair of MTAs (client and server)





Architecture of e-mail - Third Scenario:

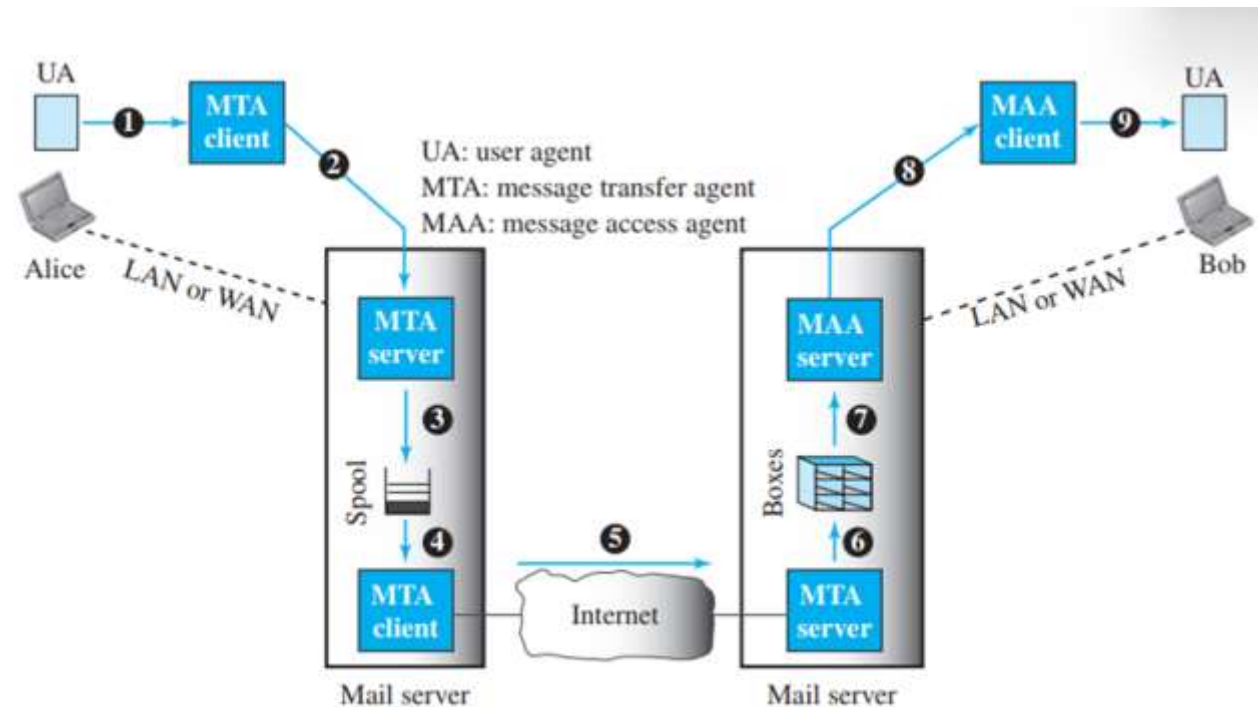
- When the sender is connected to the mail server via a LAN or a WAN, we need two UAs and two pairs of MTAs (client and server).





Architecture of e-mail - Fourth Scenario:

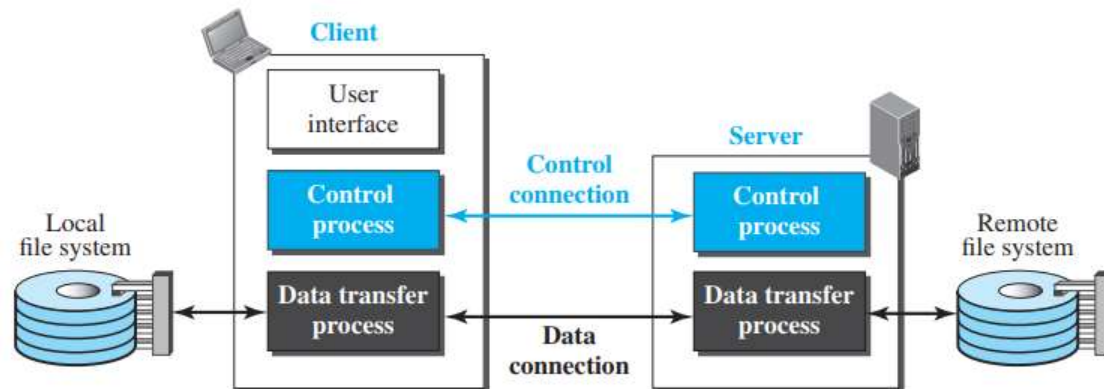
- When both sender and receiver are connected to the mail server via a LAN or a WAN, we need two UAs, two pairs of MTAs (client and server), and a pair of MAAs (client and server). This is the most common situation today





File Transfer Protocol (FTP)

- File Transfer Protocol (FTP) is the standard mechanism provided by TCP/IP for copying a file from one host to another.
- FTP differs from other client/server applications in that it establishes two connections between the hosts.
 - One connection is used for **data transfer**,
 - the other for **control information** (commands and responses)





File Transfer Protocol (FTP)

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- FTP differs from other client/server applications in that it establishes two connections between the hosts.
 - One connection is used for **data transfer**,
 - the other for **control information** (commands and responses)
- FTP uses two well-known TCP ports:
Port 21 is used for the control connection, and
port 20 is used for the data connection.



File Transfer Protocol (FTP)

- The client has three components:
 - user interface
 - client control process
 - client data transfer process
- The server has two components:
 - server control process
 - server data transfer process
- The control connection is made between the control processes.
- The data connection is made between the data transfer processes.

